Optimum Conditions for Enzymes

Enzymes=

Enzymes are made of proteins and are quite complex. They work best under certain conditions.

Effects of pH

pH means how acidic or basic conditions are.
Enzymes are affected by changes in pH. The most favorable pH value - the point where the enzyme is most active - is known as the optimum pH.

Extremely high or low pH values generally result in complete loss of activity for most enzymes.

The optimum pH value will vary greatly from one enzyme to another.

Pepsin works in the stomach where conditions are very acidic. Not surprisingly pepsin works best in low pH conditions. That means it likes to work in acid conditions.

Plot the following information onto a graph.

Activity for the Enzyme Amylase at Different pH

pH	Amylase Enzyme activity(%)
5	5
6	50
7	100
8	50
9	5

Questions.

What is the optimum pH for amylase?

So does amylase work best in acidic, basic or neutral solutions?

Effects of Temperature.

Plot the following data onto a line graph.

Temperature (^{0}C)	Enzyme activity(%)
0	0

10	20
20	45
30	90
40	95
50	60
60	20
70	10
80	0

Questions.

1. At what temperature did the enzyme have the highest activity.

2. What does the graph tell you about enzyme activity and temperature?

   If conditions of temperature get too high the enzyme's shape will change and it will no longer work. When this happens we say that the enzyme is denatured.